DRAFT — NOT SUBMISSION-READY. Generated with AI assistance from the repository's own records; every number is bound to an evidence ID in `docs/paper/sources.md`. Items marked [**UNRESOLVED**] require a decision or verification by the accountable human author(s). The final confirmatory campaign ($\geq$ 40 new tasks, `docs/PAPER_I_PLAN.md`) has not been run; this draft reports exploratory results and one pre-registered confirmatory run.

Beyond Passing Tests: Specification-Only Contract Tests Against Silent Failures of a Local Coding Agent

Oleksandr Pylypenko (Researcher) — ORCID 0009-0004-9610-9242
[UNRESOLVED: affiliation line and contact e-mail as the author wants them printed; CRediT roles]

Abstract

Coding agents are usually judged by whether they make the available tests pass. We study a local 12-billion-parameter agent (`gemma4:12b`, run on a consumer laptop GPU) that repairs small Python modules whose docstrings state more rules than the visible tests check. The agent often makes the visible tests pass, reports success, and violates a rule that only the docstring states — a *silent false success*. We evaluate a privacy-bounded division of labour: a cloud model receives only the interface of the code under repair — signatures, docstrings and data-type fields, never function bodies — and writes a unit-test module (*contract tests*) that the local agent must also satisfy; acceptance is decided by an independent hidden evaluation that neither model sees. Across 19 tasks and three seeds per task, verified completion rose from 32/57 to 48/57 runs and silent false successes fell from 25/57 to 3/57 [E5, E6]. Treating the task rather than the run as the unit of inference, 7 tasks improved and 1 worsened (sign-flip permutation $p = 0.023$; mean gain in verified rate +0.28, cluster-bootstrap 95% CI 0.09–0.49) [E7]; a pre-registered run on six fresh tasks was significant at run level ($p = 0.039$) but not at task level (3 improved, 0 worsened, $p = 0.25$) [E5, E7]. The tests must see the full interface: without data-type fields, the cloud wrote impossible tests that turned solved tasks into failures (6 losses, 0 wins, $p = 0.031$) [E3]. With the full interface, no over-constrained test was found on 47 independent valid implementations, and the contract tests killed every mutant the hidden evaluation killed, against 61.6% for the visible tests [E8]. Self-checking by the same local model, cloud guidance after failure, larger generation limits and one-step harness adaptation did not help [E2, E9, E10]. The gain costs at least 1.7–4.3× wall time [E5, E6]. The effect is concentrated in tasks whose rules live only in the docstring; whether it holds on naturally occurring bugs is the question for the planned confirmatory campaign.

1. Introduction

A passing test suite is the usual stopping criterion for an autonomous coding agent. It is also a weak one: when tests cover only part of the specification, an agent can satisfy the tests while violating the rest. Benchmark studies have shown that insufficient tests overstate the correctness of generated code — extending HumanEval's tests reduced pass@k by up to 19.3–28.9% across 26 models [R3]. For an agent that works unattended, the failure is worse than a visible error: the agent reports success, and nothing downstream objects.

We met this failure while building a hybrid orchestrator in which a cloud "architect" plans and a local model does the editing, so that source code stays on the machine. With a 12B local worker, the dominant failure on specification-heavy repair tasks was not inability to edit code

or to follow a plan: the worker made the visible tests pass and silently missed rules stated only in the docstring [E1, E2]. Generation-limit escalation, thinking on retry, cloud guidance after a failed attempt and one-step harness adaptation did not change that [E9, E10].

This paper asks a narrow question: **can tests written by a cloud model from the specification alone reduce silent failures of a local coding agent, without sending the implementation to the cloud, and at what cost?** Our contributions are:

1. A measurement of silent false successes of a local coding agent on specification-heavy repair tasks, with an independent hidden evaluation as ground truth.
2. A privacy-bounded contract-test mechanism — the cloud sees signatures, docstrings and data-type fields only — and evidence that interface completeness decides whether it helps or harms.
3. An evaluation that separates run-level from task-level inference, reports negative results for four alternative interventions, and checks the generated tests for over-constraint and mutation coverage without involving the agent.

We do not claim that generating tests with language models is new (Section 2), nor that the effect generalises beyond the task family studied here (Section 8).

2. Related work

Insufficient tests. EvalPlus extended HumanEval's test inputs about 80-fold with LLM- and mutation-based generation and found many solutions previously counted as correct to be wrong [R3]. Our hidden evaluations play the role of the stronger tests; our concern is the agent's own stopping criterion rather than benchmark scoring.

Tests generated by language models. CodeT generates tests with the same model that writes candidate solutions and ranks the solutions by agreement with those tests [R4]. AgentCoder separates a programmer agent from a test-designer agent that writes tests from the requirements [R5]. TestGen-LLM at Meta improves existing human-written test classes and keeps only candidates that pass filters for measurable improvement; in deployment 75% of its test cases built correctly, 57% passed reliably and 25% increased coverage [R2]. Closest to our setting, spec-driven test generation at Google first derives pre-conditions, post-conditions and undefined behaviour and then writes tests from them, improving bug detection on production bugs by 9.8 percentage points ($p = 0.0352$) [R1]. Our mechanism differs in where information flows: the test writer is a *different, remote* model that never sees function bodies, and the tests act as an acceptance gate for a *local* agent rather than as a test-suite product.

Agents that game tests. ImpossibleBench modifies tests so that they contradict the natural-language specification and measures how often agents nevertheless make them pass [R6]. Our failure mode is complementary: the agent does not manipulate tests; it stops when the visible tests pass and leaves unstated-in-tests rules unimplemented.

Harness adaptation. Harness^2 adapts an agent's harness — prompts, procedures, tools — without changing model weights [R7]. A one-step adaptation of our worker in this style did not improve held-out tasks (Section 6.7).

3. Problem and metrics

A task consists of a buggy module whose functions carry docstring specifications, a visible unit-test file, and a hidden evaluation that checks docstring rules the visible tests omit. For one run

of the agent let $V \in \{0, 1\}$ indicate that the task's own visible tests pass after the run, $H \in \{0, 1\}$ that the hidden evaluation passes, and $S \in \{0, 1\}$ that the agent reports success.

- **Verified completion** (primary): $V = 1 \wedge H = 1$.
- **Silent false success:** $S = 1 \wedge (V = 0 \vee H = 0)$; in practice almost always $V = 1$, $H = 0$.
- **Cost:** wall time of the agent run, local tokens and cloud tokens. The one-time generation of contract tests per task was counted in tokens but not timed in the runs reported here (Section 6.6).

V and H are evaluated after the run by the benchmark, on the task's own test modules only — generated contract tests are excluded — and the hidden evaluation is never shown to either model [E4].

4. System

Local agent. `gemma4:12b` served by Ollama 0.33.3 (model digest `4eb23ef187e2…`), with native tool calling over a workspace tool suite (read, write, patch files; run the verification command), `think` off, 16,384-token context, a generation limit of 2,048 tokens raised to 4,096 for one retry when a tool call is truncated, stage hints that name the next step and the failing test, and a self-correction loop of up to two attempts with a 300 s deadline per attempt [E4, E11]. Hardware: one NVIDIA GeForce RTX 4060 Laptop GPU (8 GB), 31 GB RAM; the model is partly offloaded to the CPU [E11].

Interface payload. For the files the subtask may modify, plus the non-test input modules they use, an AST pass emits class names, function signatures, docstrings and annotated class fields (names and types; defaults dropped). Function bodies and all other source are omitted, and the payload passes the orchestrator's privacy verifier before it leaves the machine [E12].

Contract-test writer. The cloud model receives the payload and an instruction to write one `unittest` module that checks every rule the docstrings state — results for representative and edge inputs and `ValueError` for each kind of invalid input described — and to test nothing the docstrings do not state [E12]. The returned module is statically validated (imports limited to `unittest` and the modules under test; no `open`, `exec`, `eval`, `getattr` and similar calls; no dunder attributes; a `TestCase` subclass present) and, if valid, written next to the task's own tests as `test_spec_contract.py`, where the agent's normal verification command discovers it. Tests are generated once per task and reused across seeds. The cloud backend was the Antigravity CLI (`agy` 1.4.3) with its default model, run in an empty temporary directory and instructed not to use tools [E12, E13]. **[UNRESOLVED: exact cloud model identifier was not recorded; record it for the final campaign.]**

Privacy boundary. Names, types and docstrings can themselves disclose information; we describe the mechanism as *privacy-bounded specification disclosure*, not as a formal privacy guarantee.

5. Experimental design

All runs use the same agent loop, deadline and attempt limit in each arm, three seeds (303, 404, 505) per task, and alternate the order of arms per seed [E4]. Tasks are small Python repair tasks (one or two modules) in which the visible tests cover only part of the docstring; each was checked offline so that the original code and partial fixes fail and only a complete fix passes both visible and hidden checks [E1].

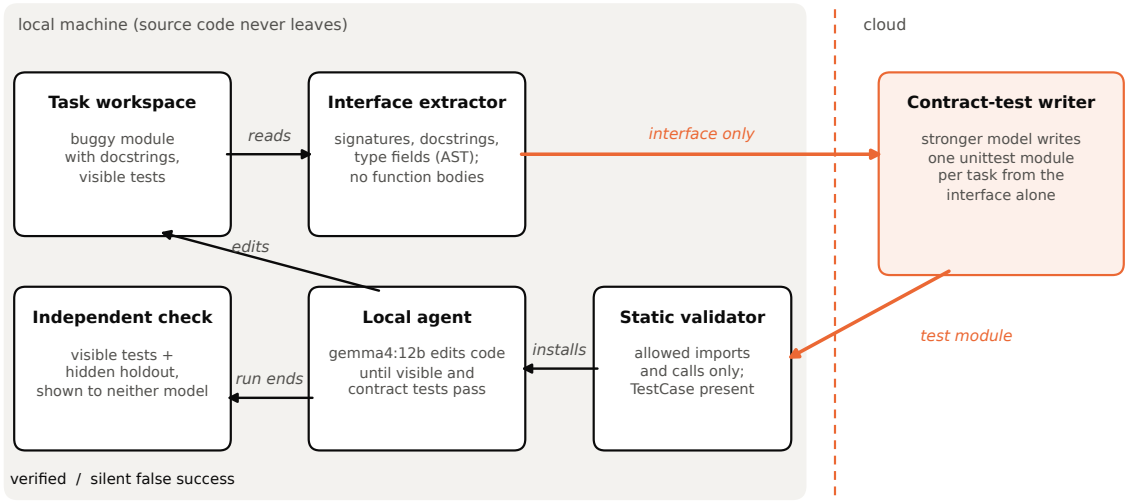


Figure 1. Information flow. Only the interface (signatures, docstrings, data-type fields) crosses the boundary to the cloud, and only a statically validated unit-test module comes back; the agent edits the code locally until its visible tests and the contract tests pass. Acceptance is decided after the run by the task's own visible tests and a hidden holdout that neither model sees. Script `scripts/analysis/paper_diagram.py` [E12, E14].

The evidence was collected in five runs (Table 1). Runs R1–R3 were exploratory; between R3 and R4 a measurement error was found and corrected (Section 6.6); R4 was pre-registered; R5 repeats R3's comparison with the corrected measurement.

Table 1. Runs.

Run	Commit	Tasks	Arms	Purpose
R1	`cc938ca`	H6, H9–H13 (6)	baseline, SPEC, SPECTESTS, ASSIST	exploratory comparison [E1]
R2	`24fbe35`	H1–H5, H7, H8 (7)	baseline, SPECTESTS v1	harm on solved tasks [E3]
R3	`750647e`	H1–H13 (13)	baseline, SPECTESTS v2	same-session comparison [E3]
R4	`7a369ac`	H14–H19 (6, fresh)	baseline, SPECTESTS v2	pre-registered confirmation [E5]
R5	`7a369ac` / `2c59e38`	H1–H13 (13)	baseline, SPECTESTS v2	corrected-measurement repeat [E6]

Baseline is the agent with stage hints and adaptive generation limit. *SPEC* gives the agent its own task's signatures and docstrings after the visible tests pass and requires one self-check of every rule. *SPECTESTS* installs cloud contract tests before the run (v1: functions' own file only; v2: plus data-type fields and dependent modules). *ASSIST* asks the cloud for step-by-step guidance after a failed attempt. Between R3 and R4 a change was made for both arms: a single stall within an attempt now leads to a fresh attempt instead of ending the subtask [E5].

Pre-registration (R4). Before any run on H14–H19 the primary endpoint (verified completion), arms, seeds, attempt limit, pairing by (task, seed) and an exact two-sided sign test were written in the project's run ledger [E5]. The ledger is a local file outside version control, so the registration is not independently time-stamped (Section 8). A first attempt of R4 crashed on a hang and used the uncorrected measurement; it was discarded and rerun unchanged except for the measurement fix [E5].

Statistics. Run-level comparisons use exact two-sided sign tests on discordant (task, seed) pairs. Because runs of one task are not independent, we also analyse the per-task difference in verified rate (SPECTESTS minus baseline, over three seeds): task-level sign test, exact sign-flip permutation test and a cluster bootstrap over tasks (20,000 resamples) [E7]. The task-level analysis was added after an external review and is post hoc with respect to R4.

6. Results

6.1 Exploratory comparison (R1)

On six specification-heavy tasks, verified completion was 7/18 for the baseline, 5/18 with local self-checks (SPEC), 16/18 with contract tests (SPECTESTS) and 8/18 with guidance after failure (ASSIST); SPECTESTS versus baseline gave 10 wins and 1 loss (run-level $p = 0.012$) [E1]. Runs that passed the visible tests but failed the hidden evaluation numbered 11 for the baseline, 10 for SPEC, 0 for SPECTESTS and 10 for ASSIST [E1]. (R1 scored the SPECTESTS arm with a stricter criterion than the baseline; Section 6.6.)

6.2 Interface completeness decides between help and harm (R2, R3)

On seven tasks the baseline already solves, the first version of the contract tests (v1) — written from the functions' own file only — reduced verified completion from 21/21 to 15/21 (0 wins, 6 losses, $p = 0.031$) [E3]. Five losses were on H1 and H3, where the data types live in other modules; grading the generated modules against reference implementations showed 19 of 20 tests wrong for H1 (a non-existent field name) and 11 of 12 for H3 (a missing constructor argument) [E3]. With data-type fields and dependent modules in the payload (v2), every generated module for the 13 tasks passed on its reference implementation, and the solved tasks lost 2 of 21 runs (not significant); on the four tasks where the baseline mostly stopped at the visible tests, verified completion rose from 2/12 to 10/12 (8 wins, 0 losses, $p = 0.0078$) [E3].

6.3 Pre-registered run on fresh tasks (R4)

On six tasks not used before, verified completion was 4/18 for the baseline and 11/18 with contract tests (8 wins, 1 loss, 9 ties; run-level $p = 0.039$), and silent false successes were 14/18 and 3/18 [E5]. At task level, three tasks improved (H14 0/3 $\rightarrow$ 3/3, H15 0/3 $\rightarrow$ 3/3, H16 0/3 $\rightarrow$ 1/3), none worsened and three were unchanged (H17 0/3 both, H18 1/3 both, H19 3/3 both): sign test $p = 0.25$, mean gain +0.39 (cluster-bootstrap 95% CI 0.06–0.72) [E5, E7]. The three remaining silent failures were all on H17; the only violated hidden rules concern empty pairs (`"a=1;;b=2"`, `"a=1;"`), a reading of the docstring that is arguably ambiguous; grading H17 without those two cases gave 7/18 versus 14/18 with the same 8 wins and 1 loss [E5]. All six generated modules passed on independent reference implementations [E5].

6.4 All 19 tasks with corrected measurement (R4 + R5)

R5 repeated the R3 comparison on H1–H13 with the corrected measurement: 28/39 versus 37/39 (10 wins, 1 loss, run-level $p = 0.012$), silent false successes 11/39 versus 0/39, and 26 of 27 SPECTESTS runs verified on the nine tasks the baseline solves 3/3 (one H6 loss) [E6]. R5 was interrupted after 36 of 78 runs by a planned shutdown and resumed from its saved results with the same generated tests [E6]. Pooled over R4 and R5 (19 tasks, identical engine and checks): verified completion 32/57 versus 48/57, silent false successes 25/57 versus 3/57 [E5, E6]. Table 2 and Figure 2 give the task-level picture; Figure 3 shows silent false successes.

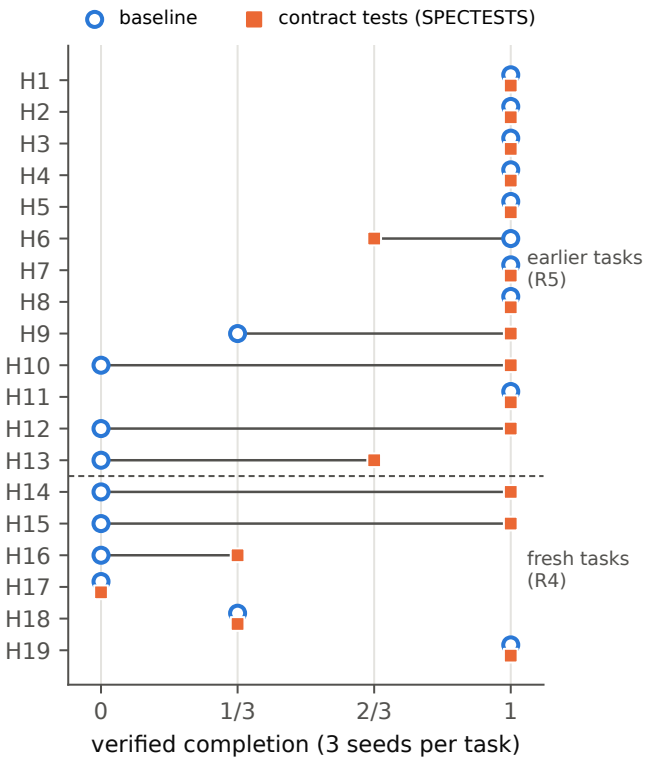


Figure 2. Verified completion per task (three seeds) for the baseline (open circles) and with contract tests (filled squares); lines join the two arms where they differ. Tasks below the dashed line are the fresh tasks of the pre-registered run R4; tasks above are from R5. Data: E5, E6; script `scripts/analysis/paper_figures.py` [E14].

Table 2. Task-level analysis (unit = task; difference in verified rate over three seeds) [E7].

Set	Better / worse / equal	Sign test	Sign-flip permutation	Mean gain (95% cluster-bootstrap CI)
Fresh H14–H19 (R4)	3 / 0 / 3	0.25	0.25	+0.39 (0.06, 0.72)
H1–H13 (R5)	4 / 1 / 8	0.38	0.13	+0.23 (0.03, 0.46)
All 19 tasks	7 / 1 / 11	0.070	0.023	+0.28 (0.09, 0.49)

The gain is concentrated in seven tasks whose rules are stated only in docstrings (H9, H10, H12, H13, H14, H15, H16); eleven tasks are unchanged and one (H6) lost one of three runs [E7].

6.5 Quality of the generated tests

The generated tests were examined without the agent [E8]. **Over-constraint.** Forty-seven distinct implementations that passed both the visible tests and the hidden evaluation in runs *without* contract tests were reconstructed from recorded diffs (initial-state hashes checked) and run against the v2 contract modules: 6 of 91 implementation–module pairs failed, all on H4 (a negative weight must raise) and H9 (a `+` sign or a non-ASCII prerelease character must be rejected). In each case the docstring states the rule and the implementation violates it; the hidden evaluation does not test it. We found no contract test that requires behaviour the docstring does not state. **Mutation coverage.** Of 448 single-point AST mutants of reference implementations

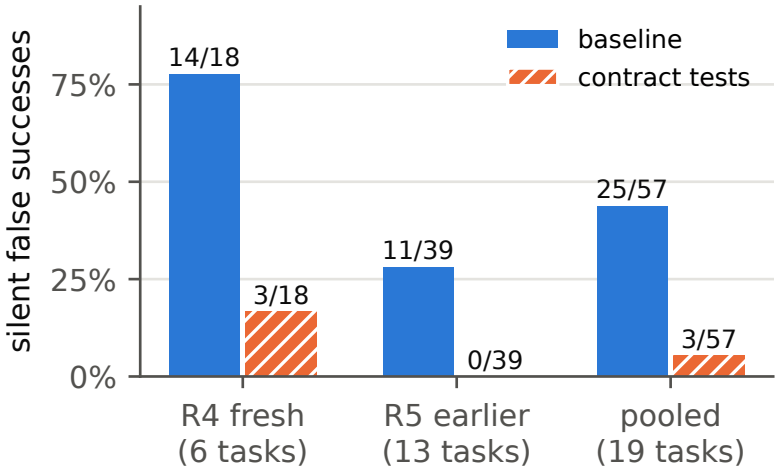


Figure 3. Silent false successes — the agent reports success but the independent check fails — per run and pooled. Data: E5, E6 [E14].

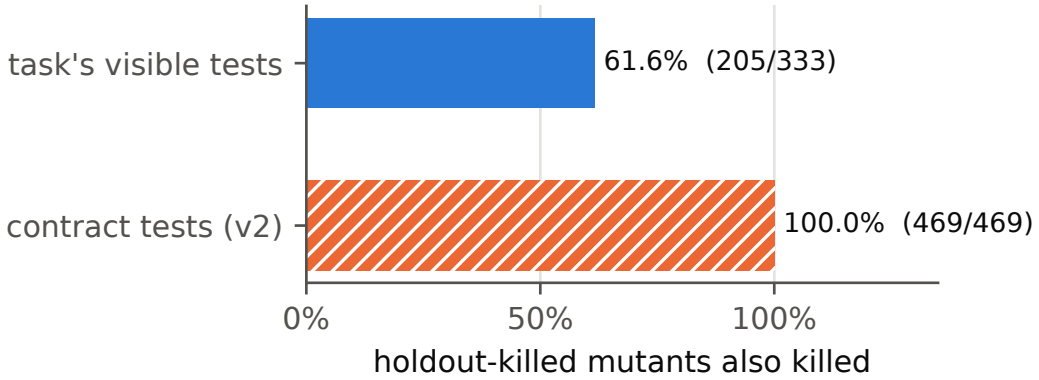


Figure 4. Of the single-point mutants that the hidden evaluation kills, the share also killed by the task's visible tests and by the v2 contract tests (contract counts are mutant–module pairs). Data: E8 [E14].

(19 tasks), the hidden evaluation killed 333; of those, the visible tests killed 205 (61.6%) and the v2 contract modules killed all (469 of 469 mutant–module pairs). (Figure 4.) A further 84 mutants survived the hidden evaluation but were killed by contract tests; each violates a docstring rule on inspection. Thirty-one mutants survived every check. For H14–H16 the reference was a holdout-passing implementation from a SPECTESTS run, because no baseline run solved those tasks. A consequence is that the hidden evaluations are weaker than the specifications, so "verified" overstates correctness in both arms; since the contract tests enforce the missed rules, this bias favours the baseline.

6.6 Measurement correction and cost

In R1–R3 the independent visible check for the SPECTESTS arm also ran the generated tests, and the hidden evaluation ran only if that check passed; SPECTESTS runs were therefore scored against a stricter criterion than baseline runs, which can only have lowered their scores [E4]. R4 and R5 use the corrected checks (task's own tests only, hidden evaluation always run, hangs counted as failures) [E4].

Contract tests make runs longer: in R4 the summed wall time was 929 s versus 4,014 s and local tokens 256,017 versus 1,171,194, plus 91,958 cloud tokens for six generation calls; in R5, 3,092 s versus 5,273 s and 614,904 versus 1,378,211 local tokens, plus about 264,000 cloud tokens for

13 tasks [E5, E6]. The baseline is fast partly because it stops as soon as the weak visible tests pass. The wall-time figures exclude the one-time cloud generation per task, which was not timed; the reported ratios (1.7–4.3×) are lower bounds on end-to-end cost [E7]. The benchmark now records generation time separately.

6.7 Interventions that did not help

- **Self-checking by the same model** (SPEC, R1): 5/18 versus 7/18; the agent's self-checks exercised only behaviour it had already implemented [E1].
- **Cloud guidance after failure** (ASSIST, R1): 8/18; guidance is triggered by a failed attempt, and the first attempt usually passed the visible tests, so it triggered once in 18 runs [E1].
- **Escalation of generation limit, thinking on retry and cloud assistance** in a two-attempt ladder: on H6–H8 the differences came from one task and were attributed to run-to-run variation [E9].
- **One-step harness adaptation** in the style of Harness^2: on held-out tasks the default harness verified 7/12 runs and the adapted harnesses 6/12, with longer total wall time (829 s versus 601 s); the standard and verifier-gated promotion rules made the same choices on every task, so the experiment does not distinguish them [E10].

7. Discussion

The results support a specific reading: on these tasks the local agent's binding constraint was *coverage of the specification by its stopping criterion*, not its ability to write the code. Executable checks derived from the docstring changed what "done" meant for the agent, and most silent failures became either verified completions or explicit failures. That the same model checking itself did not help, while a different model writing tests from the interface did, is consistent with the self-checks sharing the agent's blind spots (Section 6.7).

The failure of v1 is as instructive as the success of v2: a plausible but wrong test is worse than no test, because the agent cannot satisfy it and runs out of time. Interface completeness — here, data-type fields — was the difference between harm and benefit, which suggests that the useful question for privacy-bounded designs is not only *what may leave the machine* but *what the remote test writer must see to avoid writing impossible tests*.

The effect is not uniform. Eleven of nineteen tasks were unchanged, because the baseline either already solved them or neither arm did; the gain sits in tasks whose rules live only in docstrings. A practical deployment would therefore request contract tests selectively — when the specification states rules that the visible tests do not cover — which is also the cheaper policy given the 1.7–4.3× minimum overhead.

8. Threats to validity

- **Task selection.** All 19 tasks are small Python repair tasks written by the author; the six fresh tasks were written *after* the failure mode had been identified, so they over-represent tasks where the method helps. The effect size on naturally occurring bugs is unknown.
- **Development reuse.** Tasks H1–H13 were used while developing the mechanism; R4 is the only run on unseen tasks.

- **Statistical unit.** Seeds of one task are not independent; run-level p-values overstate generalisation to new tasks. Task-level tests (Table 2) are post hoc and, for the fresh tasks alone, not significant.
- **Pre-registration.** R4's registration lives in a local ledger outside version control and is not independently time-stamped.
- **Ground truth.** Hidden evaluations are weaker than the docstrings (Section 6.5); verified rates are upper bounds for both arms.
- **Models and hardware.** One local model, one cloud backend whose exact model version was not recorded, one consumer GPU with partial CPU offload (wall time is sensitive to background load).
- **Cost accounting.** Cloud generation time is missing from R1–R5.
- **Interrupted run.** R5 was resumed after a planned shutdown; the resumed half reused the same generated tests, but ran in a different session.
- **AI involvement.** The agent under study, the cloud test writer, and parts of the analysis and of this draft involved AI systems (Section 10); all numbers are taken from the recorded benchmark files.

9. Conclusion and next steps

On specification-heavy repair tasks, a local 12B coding agent frequently reports success after passing incomplete visible tests. Contract tests written by a cloud model from the interface alone — without function bodies — reduced such silent failures from 25 of 57 runs to 3 of 57 and raised verified completion from 32 to 48 of 57, provided the interface included data-type fields; the improvement is concentrated in about a third of the tasks and costs at least 1.7–4.3× in wall time.

The planned confirmatory campaign (`docs/PAPER_I_PLAN.md`) addresses the main threats: at least 40 new tasks including real bugs from open-source projects (power ≈ 0.87 for a task-level sign test at the observed effect distribution), a control arm in which the same cloud model receives the same interface and returns a natural-language checklist instead of tests, an ablation of how much of the interface the cloud needs, a second local model, timed end-to-end cost and a publicly time-stamped pre-registration.

10. Declarations

- **Data and code availability:** code, task definitions, raw benchmark results and analysis scripts are in the repository `cyberdid/antigravity-codex-orchestrator`, branch `feat/gemma-orion-phase2b` (results commit `fa23aac`). [**UNRESOLVED: archival DOI after the confirmatory campaign.**]
- **Use of AI:** the system under study uses `gemma4:12b` (local) and the Antigravity CLI's default cloud model; experiment orchestration, analysis scripts and this draft were produced with the assistance of Claude Code (Anthropic) and reviewed by the author(s); the draft followed the evidence-register procedure of Scientific Agent Skills [R8]. [**UNRESOLVED: wording and approval by the author(s) per venue policy.**]
- **Funding, conflicts of interest, ethics:** [**UNRESOLVED**] (no human participants or personal data are involved in the experiments).

- **Author contributions:** O. Pylypenko — sole author. **[UNRESOLVED: CRediT roles; confirm wording]**

References

R1 M. Tufano, J. McClure, J. Cambronero, S. Cheng, S. Y. Shi, R. Wei, D. Chen, F. Ivancic, L. Dalloro, P. Rondon. *Grounding AI Agents in Contracts: An Empirical Evaluation of Spec-Driven Test Generation.* SpecOps '26: 1st International Workshop on Specification-Driven Development Life Cycle, ACM, 2026. `https://research.google/pubs/grounding-ai-agents-in-contracts-an-empirical-evaluation-of-spec-driven-test-generation/` **[UNRESOLVED: DOI and pages]**

R2 N. Alshahwan, J. Chheda, A. Finegenova, B. Gokkaya, M. Harman, I. Harper, A. Marginean, S. Sengupta, E. Wang. *Automated Unit Test Improvement using Large Language Models at Meta.* FSE 2024. arXiv:2402.09171.

R3 J. Liu, C. S. Xia, Y. Wang, L. Zhang. *Is Your Code Generated by ChatGPT Really Correct? Rigorous Evaluation of Large Language Models for Code Generation.* NeurIPS 2023. arXiv:2305.01210.

R4 B. Chen, F. Zhang, A. Nguyen, D. Zan, Z. Lin, J.-G. Lou, W. Chen. *CodeT: Code Generation with Generated Tests.* ICLR 2023. arXiv:2207.10397.

R5 D. Huang, J. M. Zhang, M. Luck, Q. Bu, Y. Qing, H. Cui. *AgentCoder: Multi-Agent-based Code Generation with Iterative Testing and Optimisation.* arXiv:2312.13010v3, 2024.

R6 Z. Zhong, A. Raghunathan, N. Carlini. *ImpossibleBench: Measuring LLMs' Propensity of Exploiting Test Cases.* ICLR 2026. arXiv:2510.20270.

R7 Google Research. *Harness*[2] (software repository). `https://github.com/google-research/harness2` **[UNRESOLVED: preferred citation from the repository]**

R8 T. Kassis, V. Agarwal, Y. He, D. Patel, A. M. Brueckner. *Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents.* arXiv:2609.00065, 2026. `https://doi.org/10.48550/arXiv.2609.00065`